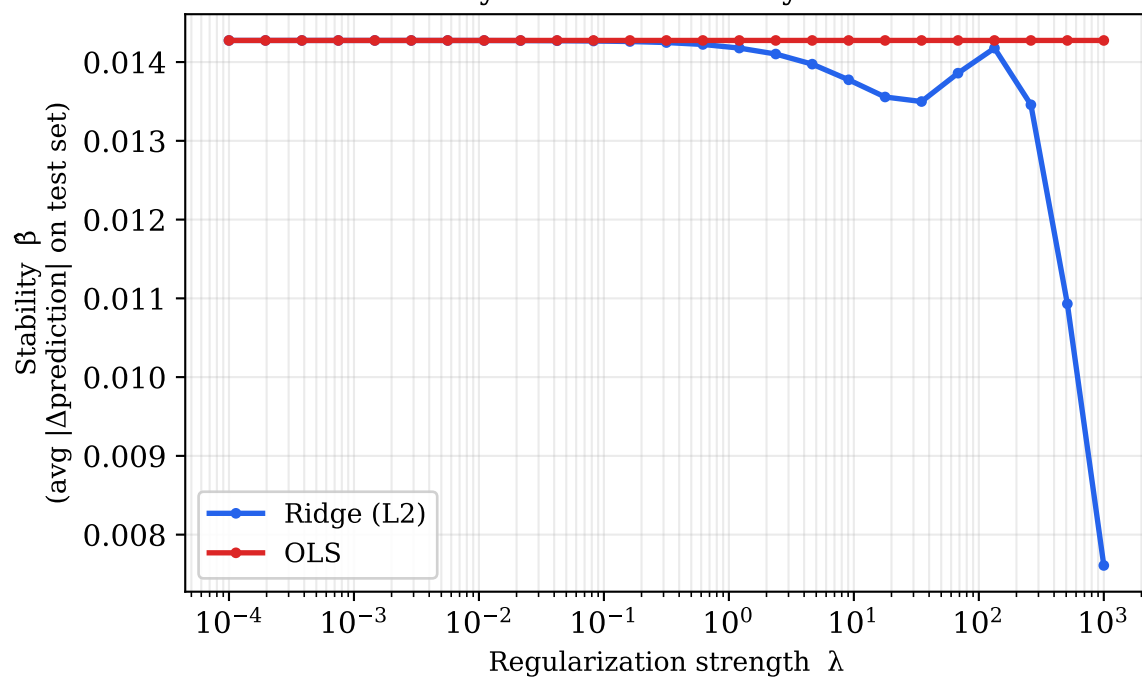


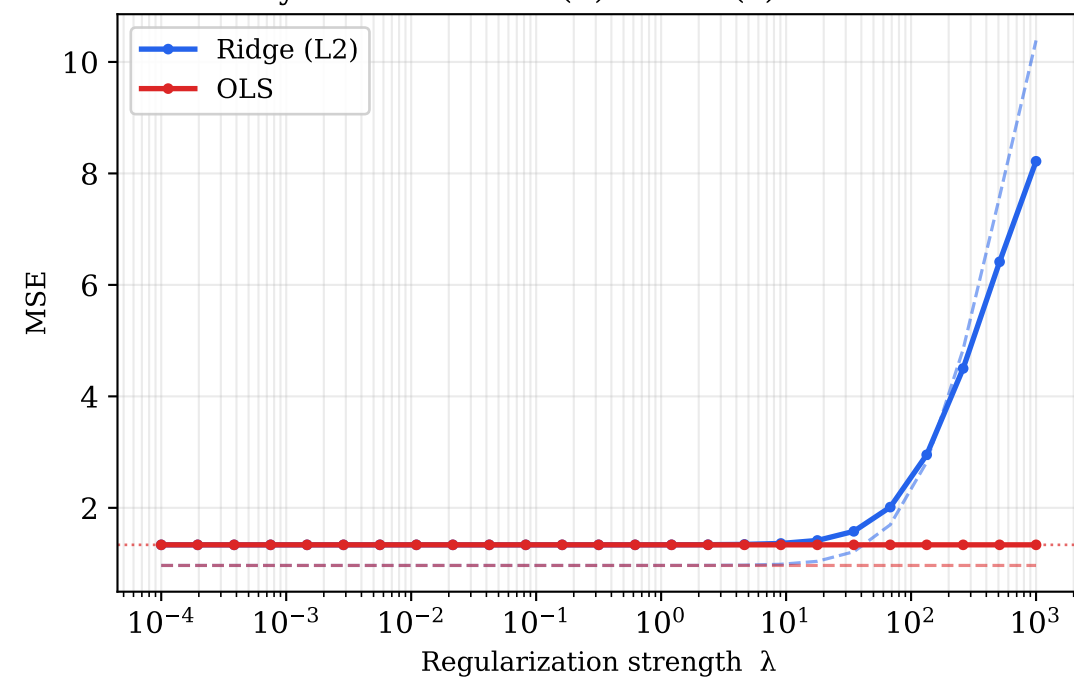
# Assignment 3 — Regularization and Stability

## Bousquet & Elisseeff (JMLR, 2002)

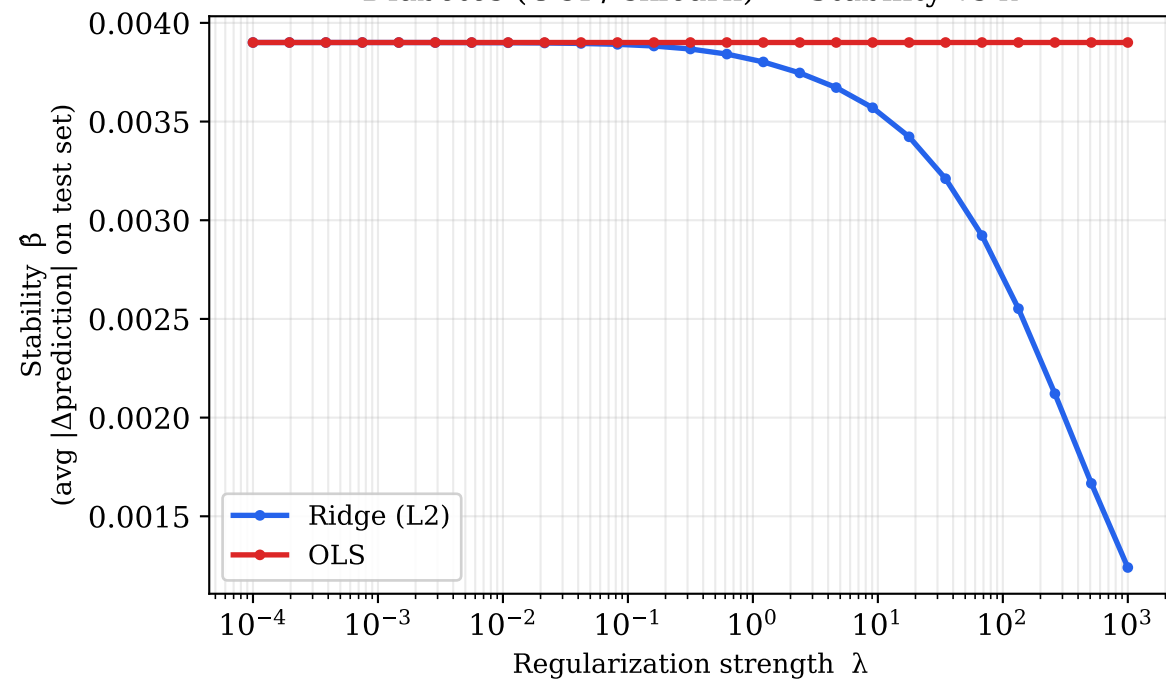
Synthetic — Stability vs  $\lambda$



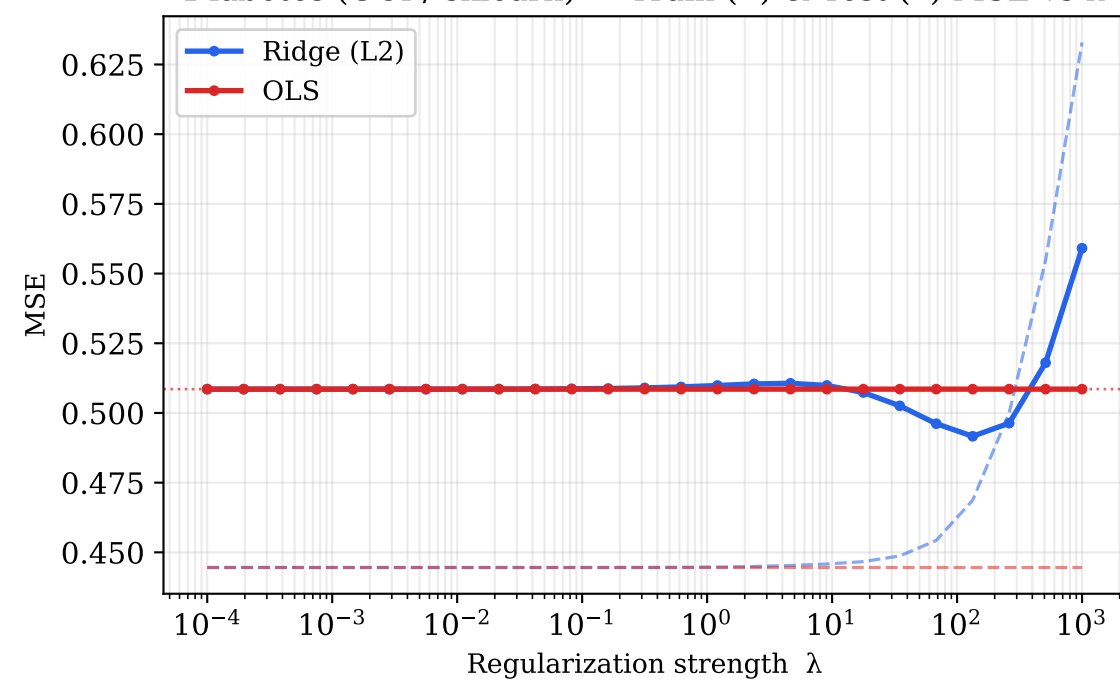
Synthetic — Train (--) & Test (—) MSE vs  $\lambda$



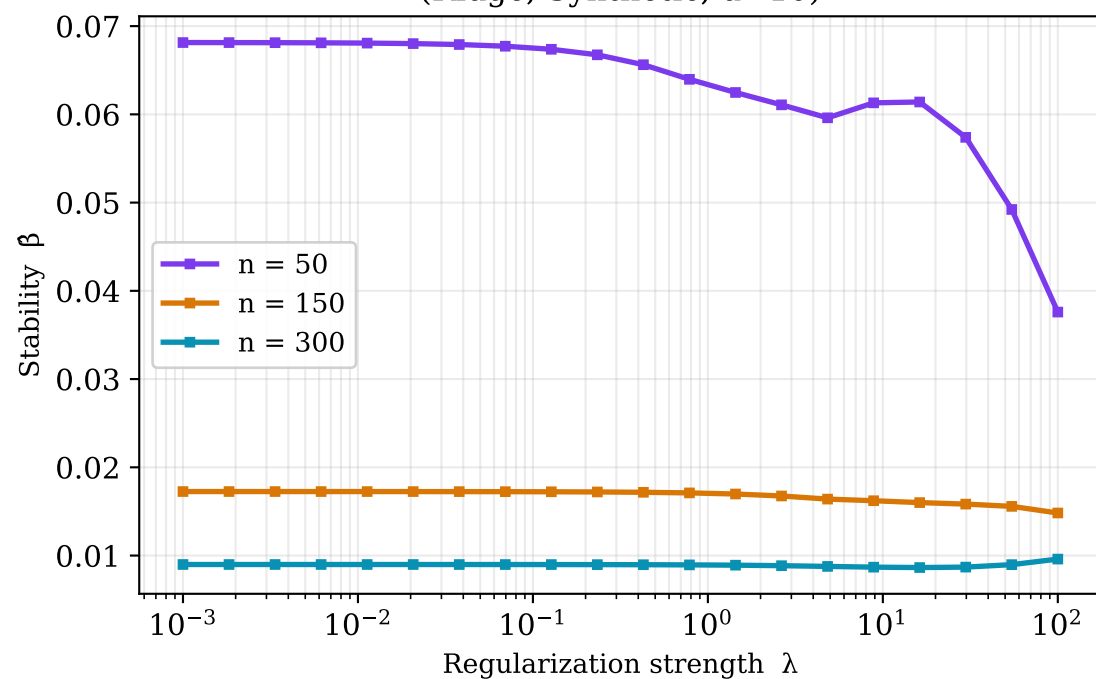
Diabetes (UCI / sklearn) — Stability vs  $\lambda$



Diabetes (UCI / sklearn) — Train (--) & Test (—) MSE vs  $\lambda$



Effect of Training Set Size on Stability  
(Ridge, Synthetic,  $d=10$ )



L1 vs L2 Stability Comparison  
(Synthetic Dataset)

